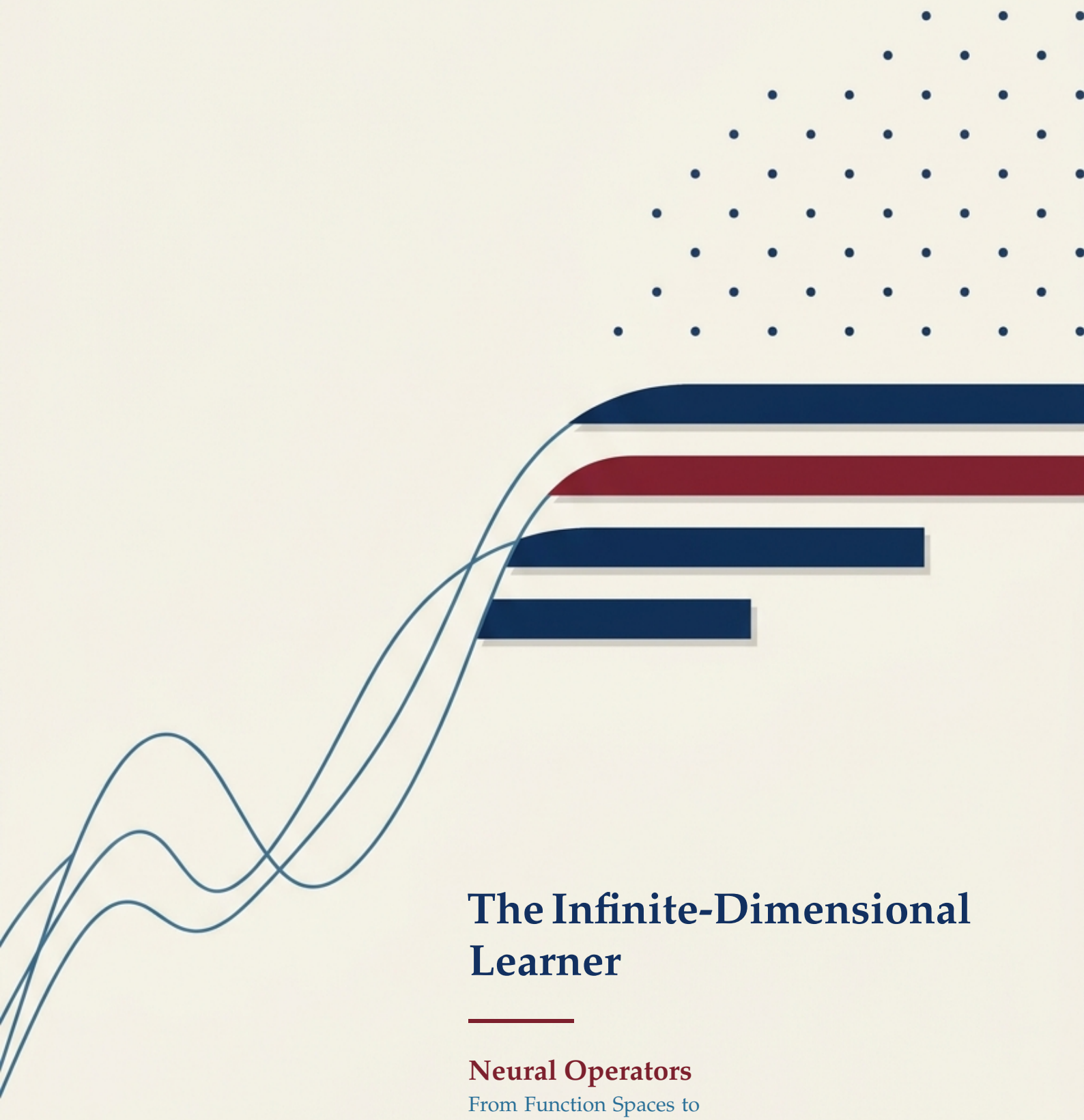


The Infinite-Dimensional Learner

Neural Operators

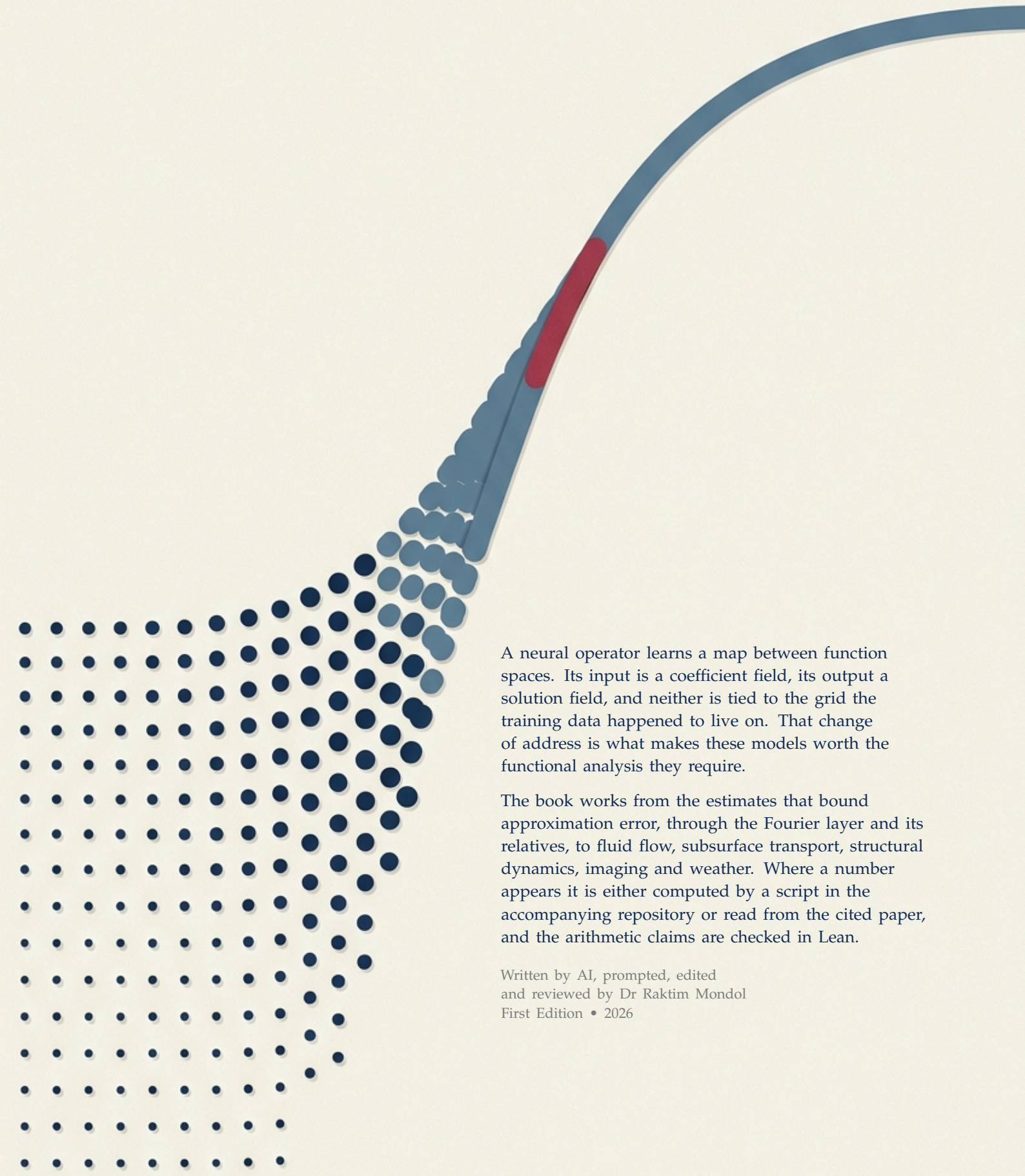
From Function Spaces to
Scientific Machine Learning

Dr Raktim Mondol

First Edition • 2026

The Infinite-Dimensional Learner

Dr Raktim Mondol



A neural operator learns a map between function spaces. Its input is a coefficient field, its output a solution field, and neither is tied to the grid the training data happened to live on. That change of address is what makes these models worth the functional analysis they require.

The book works from the estimates that bound approximation error, through the Fourier layer and its relatives, to fluid flow, subsurface transport, structural dynamics, imaging and weather. Where a number appears it is either computed by a script in the accompanying repository or read from the cited paper, and the arithmetic claims are checked in Lean.

Written by AI, prompted, edited
and reviewed by Dr Raktim Mondol
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